

Retail Sales Performance & Forecast

Prepared for the Head of Supply Chain & CFO | 4-Year Sales Analysis (2015–2018) | Retail Pulse Analytics

Executive Summary

This report reviews four years of company sales data (2015–2018) to understand what is selling, what is driving swings in demand, and what to expect over the next three months. The underlying business has grown steadily year over year, with a strong and predictable jump in sales every fourth quarter around the holidays. Our best-performing forecasting model expects total monthly sales to settle in the mid-to-high \$80,000s over the next quarter, with Office Supplies and Furniture growing the fastest while Technology holds roughly flat. We also identified a small number of unusual sales weeks worth a closer look, and grouped products into four demand types to guide how much stock to hold of each. Three practical, data-backed actions are recommended to reduce stockouts, cut wasted inventory, and prepare for the next holiday peak.

Key Findings

- Revenue by category:** Technology is the top-selling category over the four years (\$827K total), just ahead of Furniture (\$729K) and Office Supplies (\$705K) — a fairly even three-way split.
- Most consistent growth:** the East region shows the steadiest, most consistent growth over the four-year period.
- Shipping performance:** orders ship in about 4 days on average, and this is consistent across all four regions (3.9–4.1 days) — shipping speed is not a regional problem.
- Seasonality:** September, November and December are consistently the strongest sales months every year, while February is consistently the weakest — a clear, repeatable holiday pattern.
- Best forecasting model:** three forecasting methods were tested (SARIMA, Prophet, and Gradient Boosting/XGBoost). The Gradient Boosting model was the most reliable, with an average error of about 13%, versus 20–22% for the other two.

3-Month Sales Forecast (Plain Language)

Using our most accurate model, total company sales are expected to be roughly \$85,000 in Month 1, \$87,000 in Month 2, and \$87,000 in Month 3. Because no model is perfect, we cross-checked this against two other methods — the table below shows the full range they suggest. The further out the forecast, the more the models agree, which is a good sign for planning Month 2 and 3 more confidently than Month 1.

Month	Most Likely Sales	Realistic Range (all 3 models)
Month 1	~\$85,000	\$51,000 – \$85,000
Month 2	~\$87,000	\$87,000 – \$91,000
Month 3	~\$87,000	\$87,000 – \$97,000

Top 3 Anomalies Detected

- 1. Week of March 22, 2015 (largest anomaly):** sales spiked to roughly \$37,700 in one week — over five times a typical week. Both detection methods flagged this independently, giving high confidence it's a real event, most likely a large bulk or corporate order, or a quarter-end sales push.
- 2. Week of January 4, 2015 (unusually quiet):** sales dropped to about \$300 for the week — far below normal. This lines up with the typical post-holiday lull that follows the New Year sales surge.
- 3. Week of February 22, 2015 (unusually quiet):** another unusually quiet week (~\$225 in sales). This falls outside the normal holiday cycle and is worth checking against marketing activity or possible stock shortages during that period.

In total, one detection method flagged 11 unusual weeks and a second, stricter method flagged 6; both agreed on the March 22 spike, which is why it is treated as the highest-confidence anomaly.

Product Demand Segmentation & Stocking Strategy

Product sub-categories were grouped into four demand segments based on sales volume, order size, and volatility. Recommended stocking approach per segment:

Segment	Recommended Stocking Strategy
High Volume, Stable Demand	Maintain safety stock with automated replenishment; lock in long-term supplier contracts.
Growing Demand	Increase inventory and warehouse allocation; tighten forecasting ahead of peak season.
Low Volume, High Volatility	Keep minimal inventory; favor on-demand purchasing or drop-shipping.
Declining Demand	Slow down reordering; use discounting and bundling to clear existing stock.

Business Recommendations

- **1. Increase Q4 inventory for the top three categories.** Build up stock and staffing for Technology, Furniture, and Office Supplies ahead of September–December every year. Data backing: these three months post the highest sales of the year, consistently, across all four years analyzed — this is the single strongest, most repeatable pattern in the data.
- **2. Shift growth investment toward Office Supplies and Furniture.** Direct incremental buying budget toward Office Supplies and Furniture, which are forecast to grow roughly 50% or more over the next quarter (Office Supplies: ~\$24K → ~\$37K; Furniture: ~\$15K → ~\$36K), while Technology is projected to stay roughly flat (~\$23K). Growth dollars will go further in the categories already trending up.
- **3. Move to segment-based inventory policy.** Replace the current uniform reorder policy with segment-specific rules — safety stock for stable/high-volume items, drop-shipping for volatile/low-volume items, and clearance pricing for declining items. Data backing: the four demand segments differ sharply in sales volatility, so a single stocking rule either over-invests in unpredictable, low-volume items or under-stocks reliable, high-volume ones.

Risk & Limitation to Be Aware Of

This forecast relies solely on four years of historical order data (2015–2018) and does not account for live signals such as new promotions, competitor moves, or supply disruptions, so accuracy will drop around unusual events. Even the best model is off by about 13% on average, so treat forecasts as a planning range, not an exact number. The four product-demand segments were also generated by an automated algorithm; they are directionally useful but should be spot-checked by category managers before being used to set firm stocking commitments.

Source: RetailPulse sales analysis (Superstore dataset, 2015–2018) · Live dashboard: retailpulse-rgq7dm3dok9ap7bhmapdcu.streamlit.app